

Q1.) What are the different protocols you observe at following layers of the Protocol Stack?

- (A) Application layer
- (B) Transport layer
- (C) Network layer

Sol<sup>n</sup>: Application layer protocols:

- HTTP
- DNS
- TLS v1.2
- TLS v1.3
- MDNS
- LLNMR
- PKIX-C

(B) Transport layer protocol used:

- UDP
- TCP
- QUIC

(C) Network layer protocol used:

- ICMPv6

Q.2.) What is the total amount of data being received for following two cases?

A. When you access <http://neverssl.com/>

B. When you access <https://www.cornell.edu/>

(A) Ans: ~~0 KB~~ 0 bytes

(B) Ans: ~~0 KB~~ 0 bytes

Q.3) Answer when you access the site <http://neverssl.com/>.

(A) How many HTTP GET requests do you observe? List down the GET requests.

Soln: 333 (i) ~~GET/we~~  
 (ii) GET/online HTTP/1.1  
 (iii) GET/online HTTP/1.1  
 (iv) GET/favicon.ico HTTP/1.1

(B) What version of HTTP the server is running?

Soln: HTTP/1.1

(C) For each of the HTTP GET requests, find out:

(i) The total number of TCP segments being received.

Ans: (i) GET/online HTTP/1.1 → 1  
 (ii) GET/online HTTP/1.1 → 2  
 (iii) GET/favicon.ico HTTP/1.1 → 1

(ii) The total amount of data being received in the corresponding HTTP response message:

Ans: Data received  
 (i) GET /online HTTP/1.1 → 303 bytes  
 (ii) GET /online HTTP/1.1 → 1519 bytes  
 (iii) GET /favicon.ico HTTP/1.1 → 124 bytes

- (D) Identify TCP three way handshake used to establish the connection with the http://neverssl.com. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

Soln:

| Handshake Packet | Frame No. | Sequence No. | Acknowledgment No. |
|------------------|-----------|--------------|--------------------|
| (i) SYN          | 3121      | 0            | — (N/A)            |
| (ii) SYN-ACK     | 3260      | 0            | 1                  |
| (iii) ACK        | 3262      | 1            | 1                  |